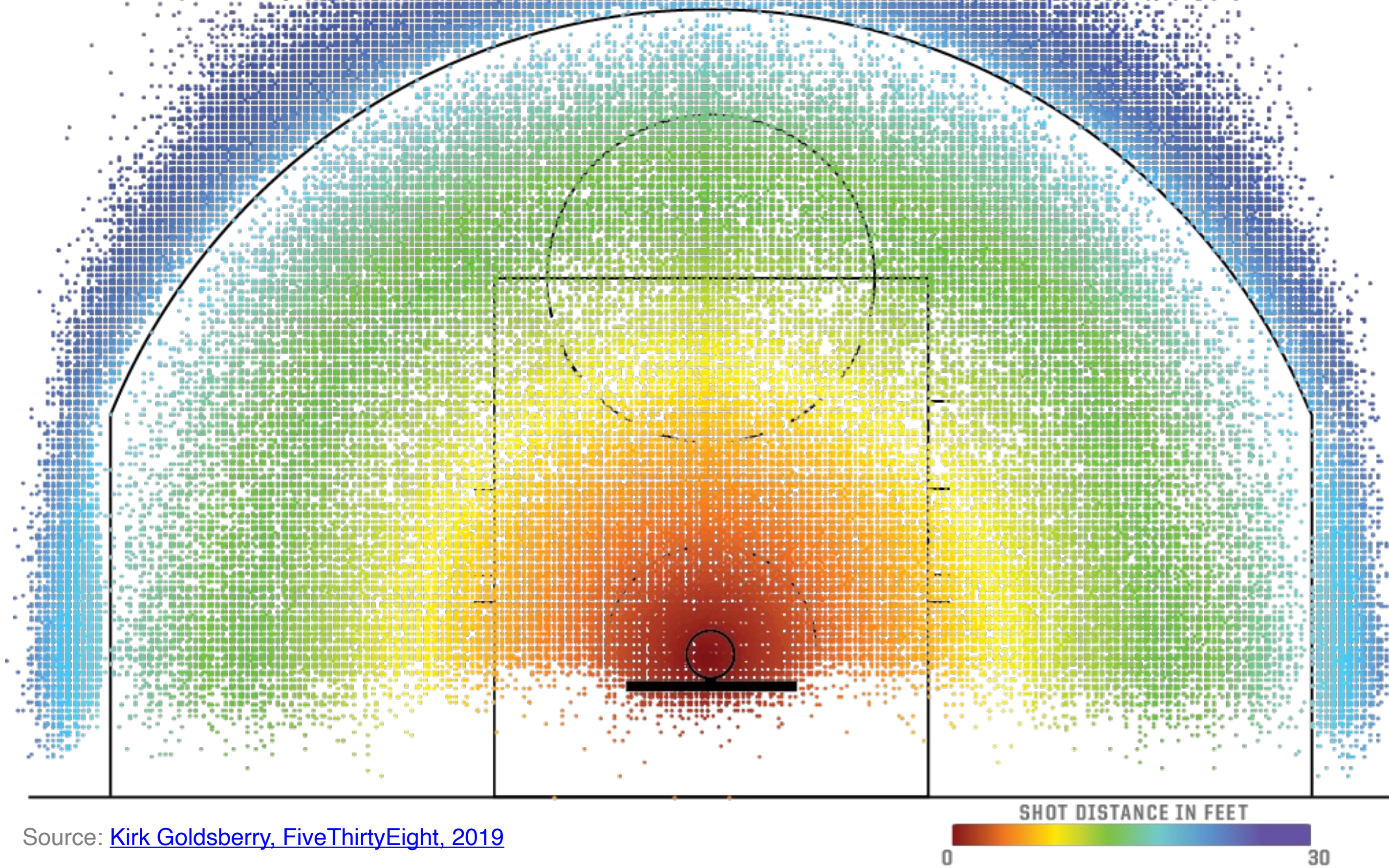


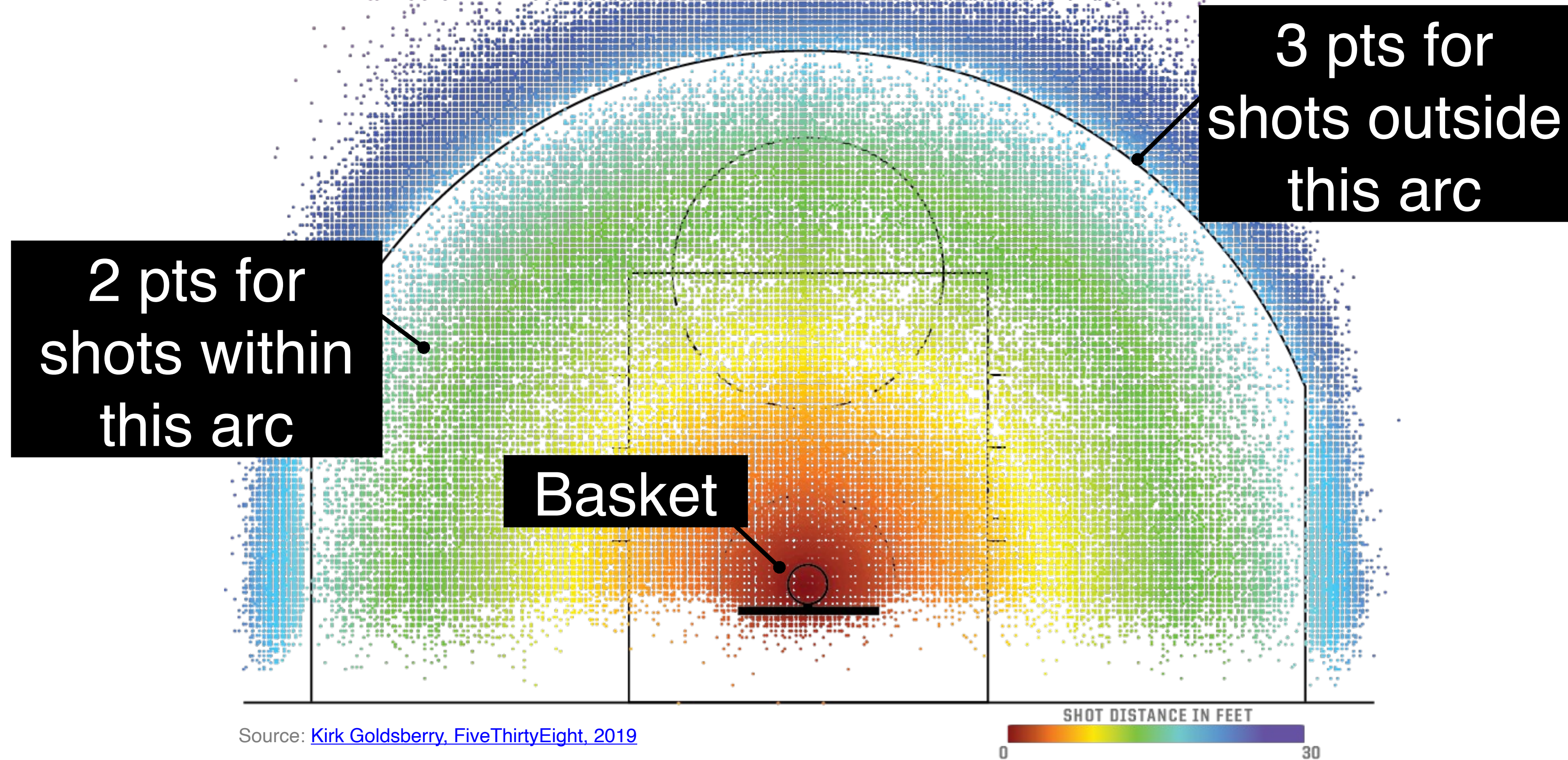
**Is the game of
basketball broken?**

NBA SHOT LOCATIONS 2014-15



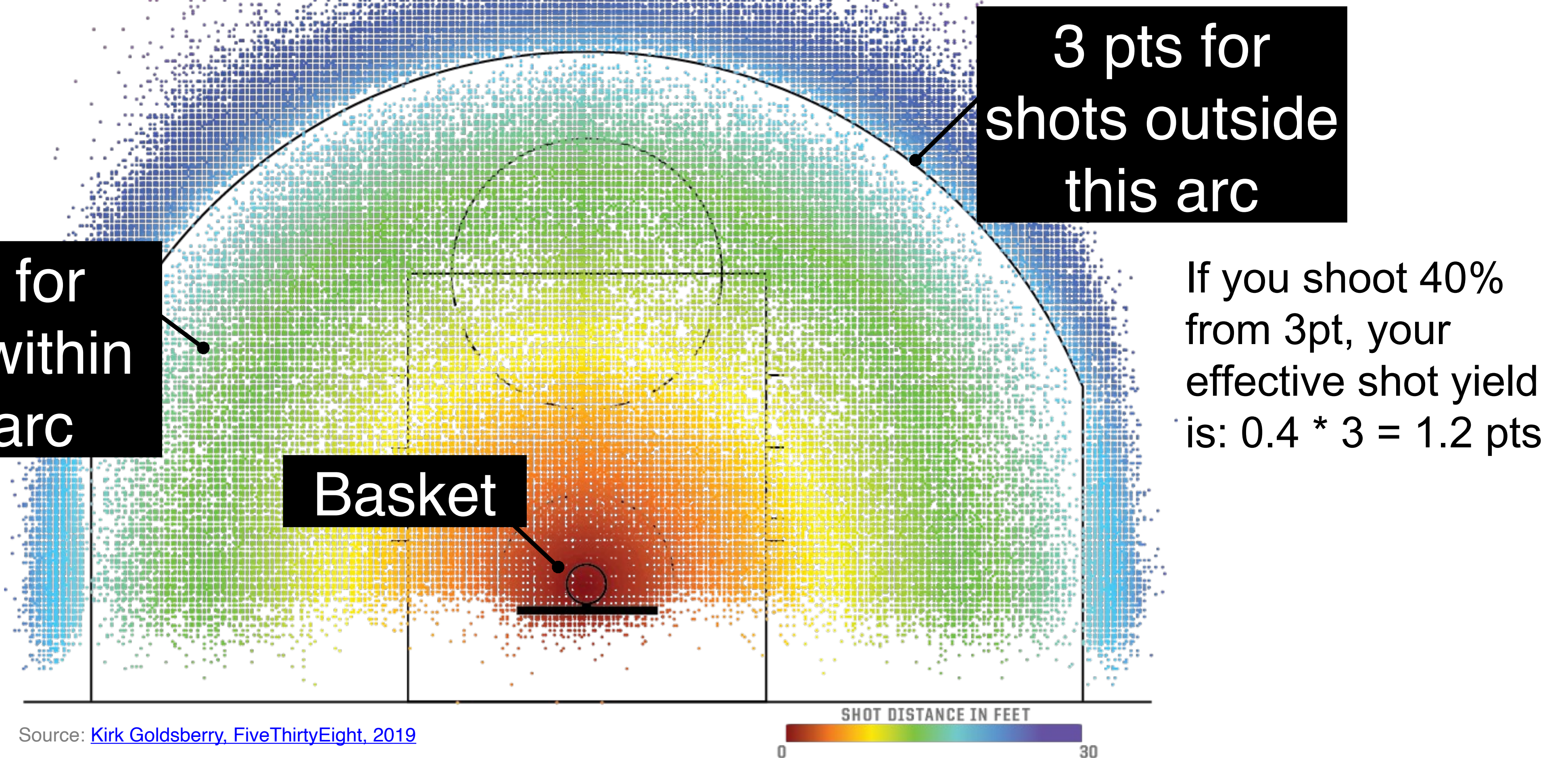
Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)

NBA SHOT LOCATIONS 2014-15



Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)

NBA SHOT LOCATIONS 2014-15

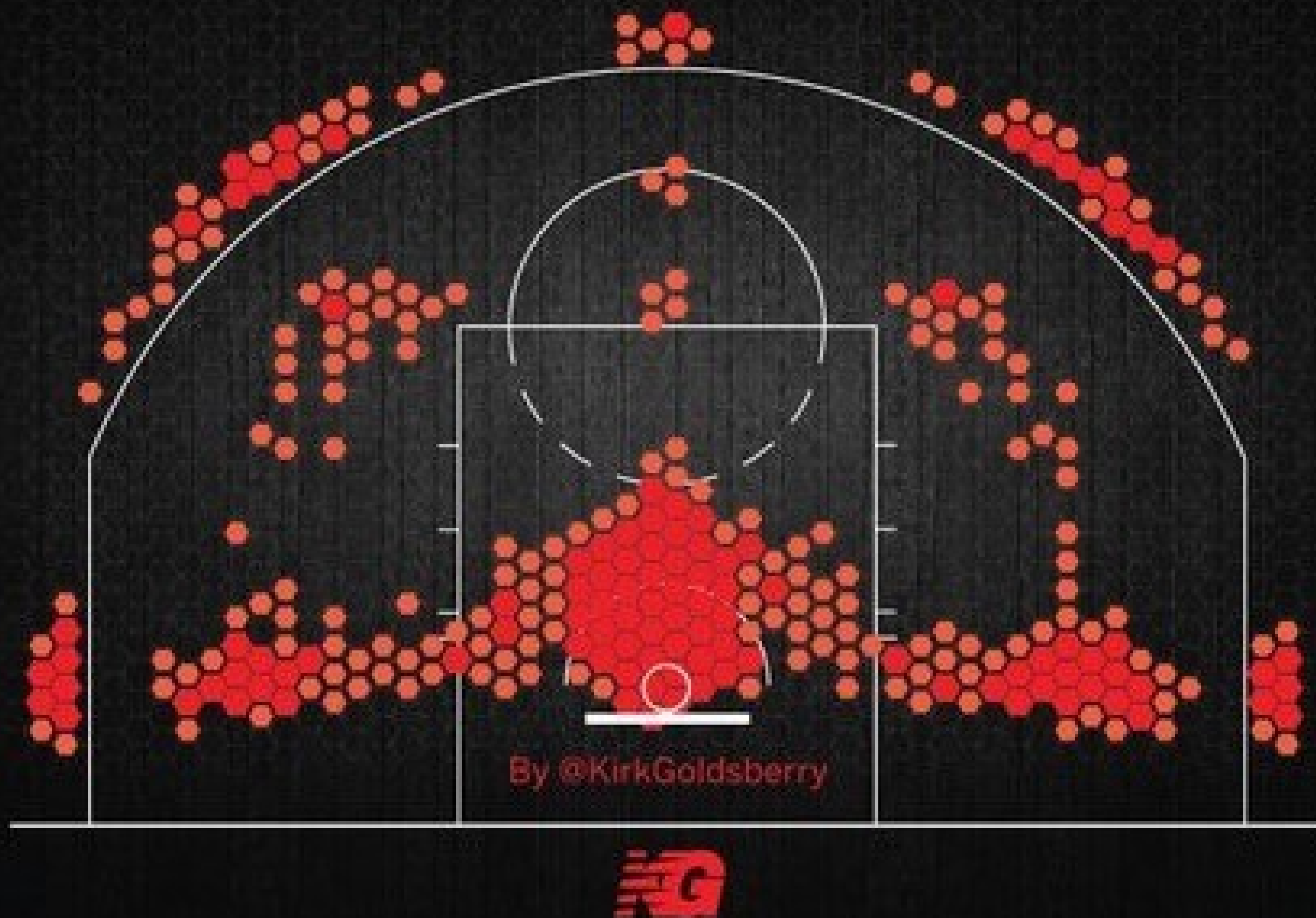


Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)

If you shoot 50% from 2pt, your effective shot yield is: $0.5 * 2 = 1$ pt

Top 200 shot locations in the NBA, 2001-02 versus 2019-20

2001-02

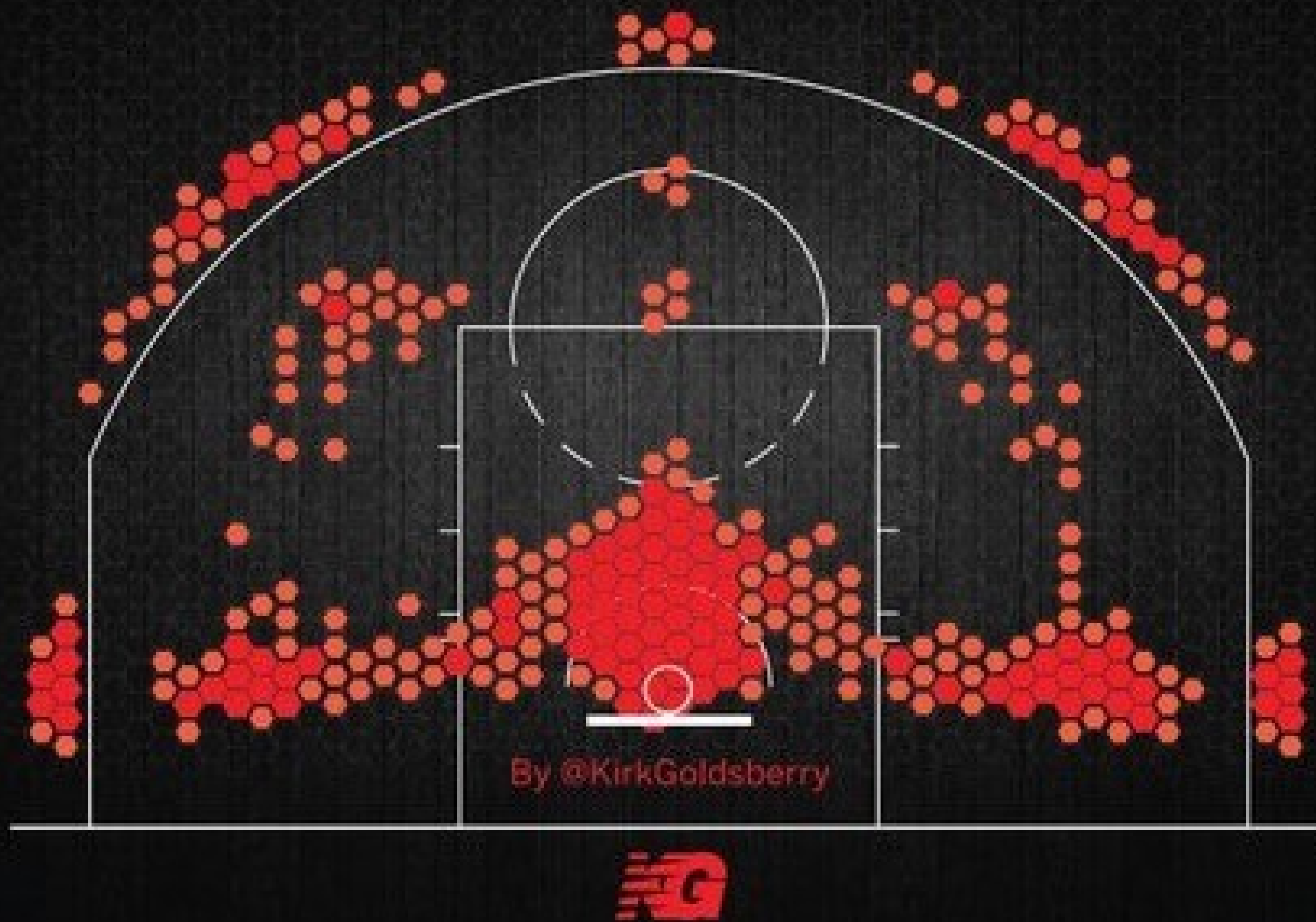


2019-20

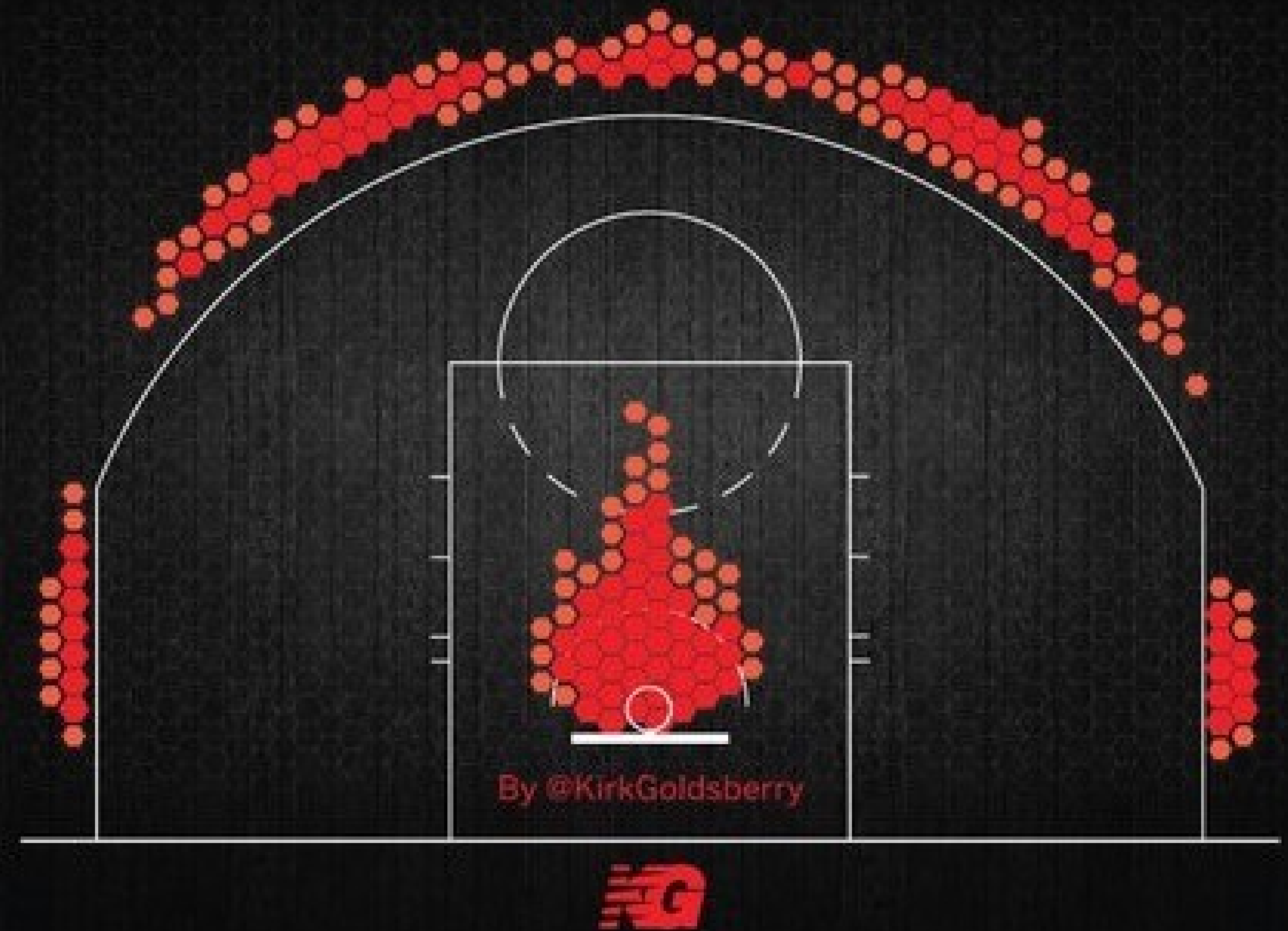


Top 200 shot locations in the NBA, 2001-02 versus 2019-20

2001-02



2019-20



Project 1: Exploratory Data Analysis (EDA)

- We will discuss what this entails over the next few weeks
- Checkpoint #1: Topic and dataset picked: **by Sept 13 @ midnight**
- Checkpoint # 2: Questions to explore picked: **by Sept 27 @ midnight**
- Final **lightning talk** due: **Oct 11 @ midnight**
- Final **notebook** due: **Oct 14 @ midnight**
- For now, you should pick one of your interests (research or hobby!) and think of questions you have about that interest

Coffee:

- Is Charlottesville coffee overpriced?
- What makes for a good cup of coffee? (CQI)
- Do coffee consumers tend to prefer light or dark roast?
- How does the local vs global economy affect coffee prices and sales?

Public transit:

- How often are Charlottesville buses late?
- How much overlap is there between UTS and CAT?
- How has ridership changed since COVID? Was the decision to make rides free a good one?

UVA DS2003: Communicating with Data
Fall 2026

Visualization Design:

Data Types, Tables, and Models

Sarah Groves

Adapted from Hannah Bako and Rafael Alvarado

What you will learn today

- **Data components:** item, attribute, relationship
- **Dataset Types:** Multivariate, Network/Tree
- **Data Tables:** Messy vs. Tidy Data
- **Attribute Types:** Nominal, Ordinal, Quantitative (Interval & Ratio)

The Big Picture



The Big Picture



Physical Data Types

int, float, string

Conceptual Data Types

temperature, location

Visual Channels

x, y, color, opacity

Graphical Marks

rect, line, point, area

What is data?

Which are examples of data?

6

6 feet

height = 6 feet

Bob's height = 6 feet

All are sometimes called "data"

But only the last item is genuine *datum* —
an elementary assertion of a fact

The Elementary Structure of Data

A **datum** is always an assertion of **something about something**

[Bob] [has height] [6 feet] → A, B, C

A is some **unique thing** in the world defined by a name (ID)

B is some property, defined by a **category**, connected to the thing

C is some **value** in the form of a data type (numeric or categorical) or a reference to another datum

Data always has this basic **triadic structure**

In other words, **a datum is a sentence** (proposition)

Semantic Triples

This triadic structure is called a "**semantic triple**"

It represents the most basic kind of "**fact**"

Asserting something about something

In the form of a simple **sentence**

In the form of a **subject-predicate-object** (SPO) expression

Subject: An entity or resource that the statement is **about** (e.g., "Bob").

Predicate: The property or relationship **linking** the subject and object (e.g., "has a height of").

Object: The **value** or related entity (e.g., "6ft").

Data Components

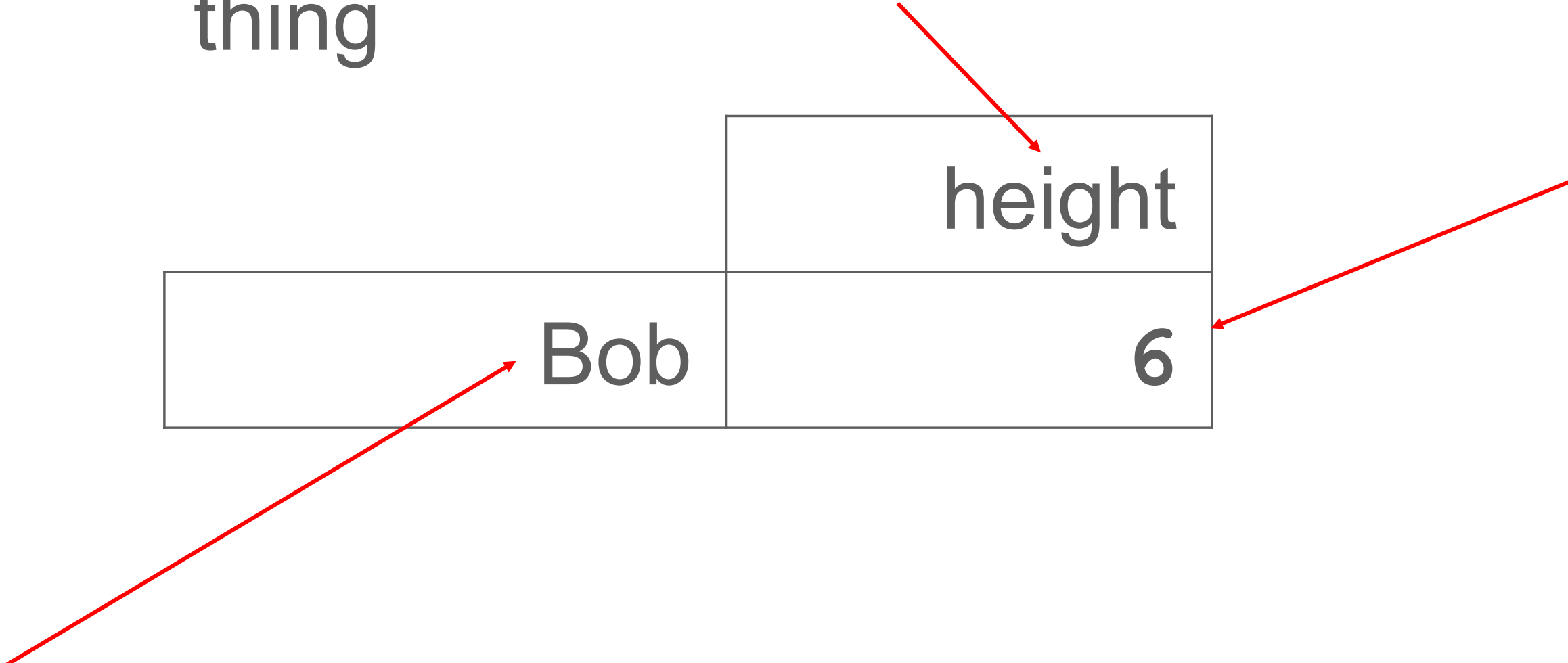
- **Items** aka data objects, data cases
(cars, employees, cities, countries, ...)
- **Relationships** aka links
(parent of, owned by, have collaborated, ...)
- **Attributes** aka data variables, fields
(mpg, age, population, GDP, ...)

Dataset Types

- **Multivariate Data: Items and Attributes**
- Networks & Trees

You are used to seeing a semantic triple in this form —
as the building block of a **table**

P is a **column/attribute** that
stands for a feature of the the
thing



	height
Bob	6

O is a **cell** that
contains some
value in the form of
a data type
(numeric or
categorical) or a
reference to another
datum

S is a **row/item** that stands for an
observation, some unique thing or outcome
in the world, defined by name (IDs)

When we combine data units we get a table

	height	weight
Bob	6	200
Maria	5 . 5	130

So a table is actually a bunch of sentences,
represented a certain way:

- “Bob has height 6 feet.”
- “Bob has weight 200 pounds.”
- “Maria has height 5.5 feet.”
- “Maria has weight 130 pounds.”

We can think of this as a basic form of visualization

Sentences → Table

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box	attribute	7/17/07
32	7/16/07	2-High	Medium Box		7/18/07
32	7/16/07	2-High	Medium Box		7/18/07
35	10/23/07	4-Not Specified	Wrap Bag		10/24/07
35	10/23/07	4-Not Specified	Small Box		10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	item	5 4-Not Specified	Small Pack	0.44	6/6/05
69		5 4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05
97	1/29/06	3-Medium	Small Box	0.38	1/30/06
129	11/19/08	5-Low	Small Box	0.37	11/28/08

Dataset Types

- Multivariate Data: Items and Attributes
- **Networks & Trees: Items, Attributes, and Relationships**

Semantic Triples for relationships: networks and trees

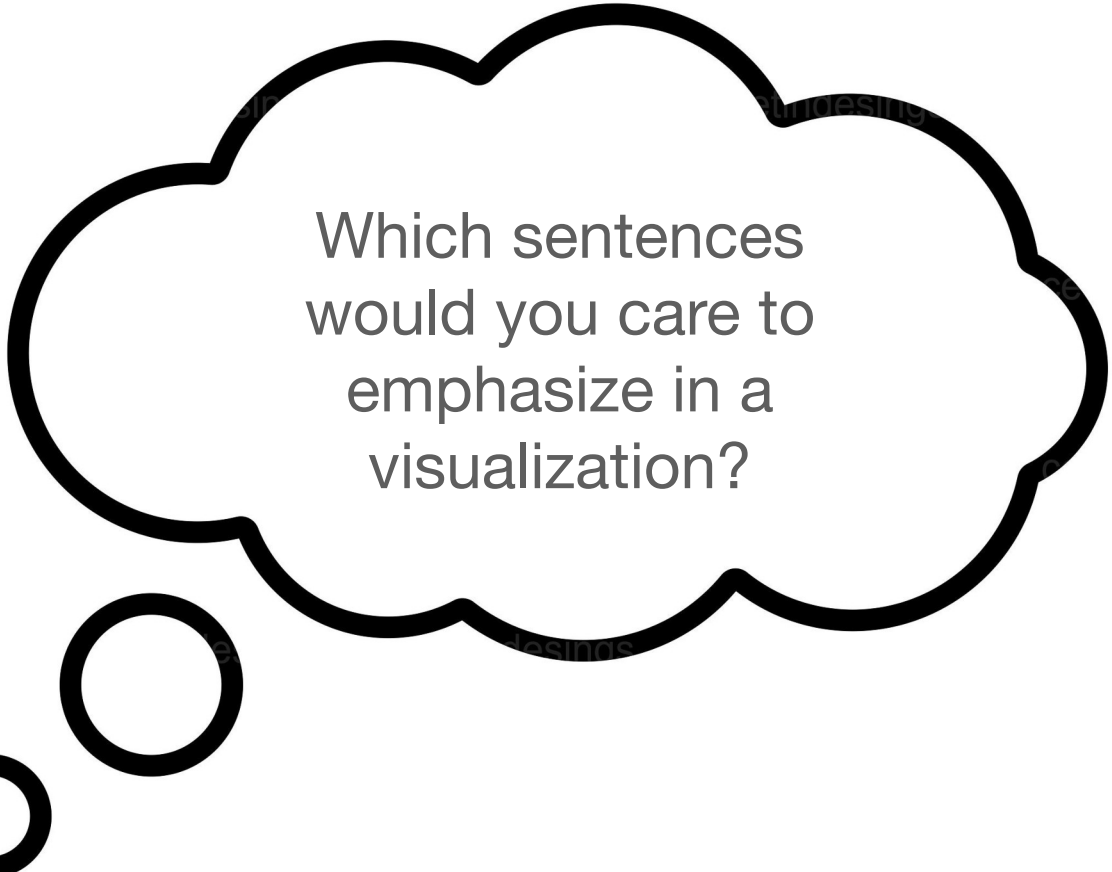
In the form of a **subject-predicate-object** (SPO) expression

Subject: An entity or resource that the statement is **about** (e.g., "Bob").

Predicate: The property or relationship **linking** the subject and object (e.g., "is the parent of").

Object: The **related entity** (e.g., "Charlie").

For network or tree data, the link is usually a *relationship*



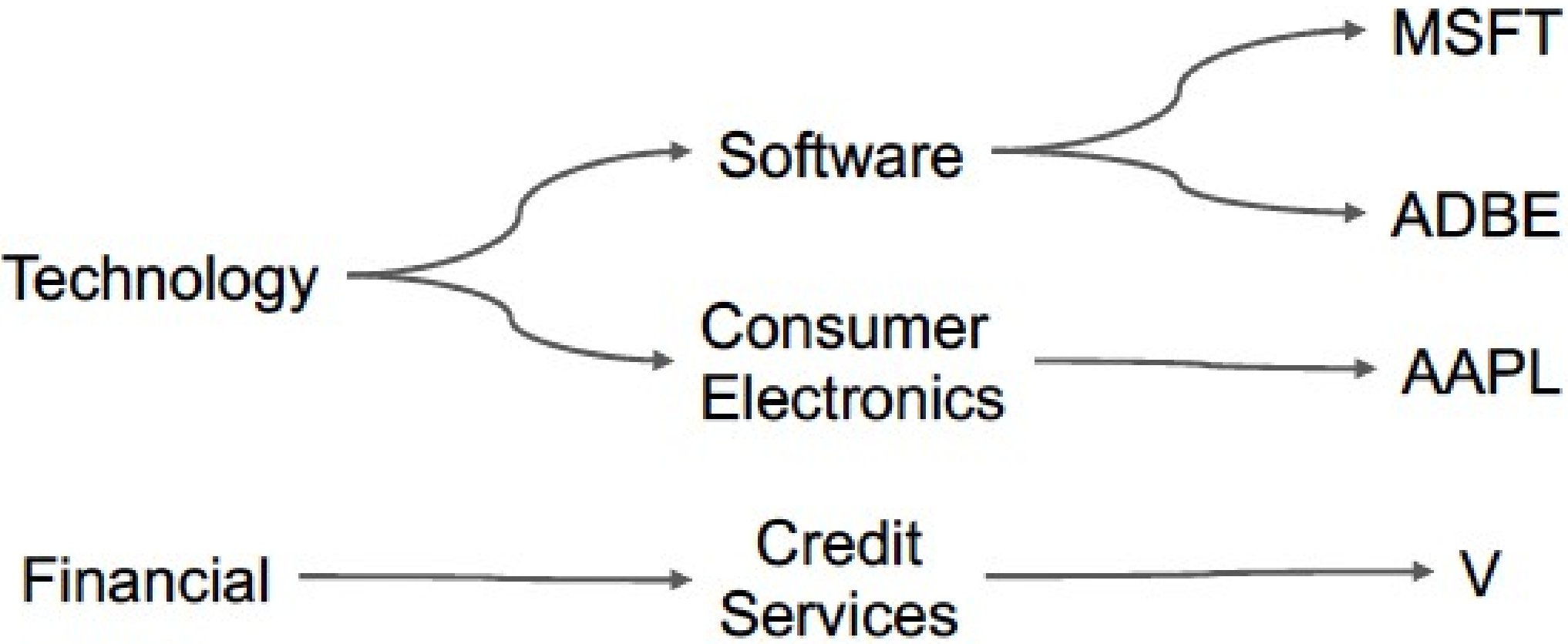
Item	Ticker	Sector	Industry	Attribute Change	Value
	MSFT	Technology	Software	- 1.40%	214.25
	ADBE	Technology	Software	- 3.12%	491.94
	AAPL	Technology	Consumer Electronics	+0.07%	120.96
	V	Financial	Credit Services	- 2.06%	204.66

What relationships exist in this table?

- ADBE is in the technology sector
- AAPL is in the Consumer electronics industry
- V experienced change of -2.06%
- MSFT and ADBE are in the same industry

Item

Attribute				
Ticker	Sector	Industry	Change	Value
MSFT	Technology	Software	- 1.40%	214.25
ADBE	Technology	Software	- 3.12%	491.94
AAPL	Technology	Consumer Electronics	+0.07%	120.96
V	Financial	Credit Services	- 2.06%	204.66





A special kind of table
in data science
is what we will call
the **standard data table**

(sometimes called "tidy data")

Anatomy of a Data Table

Row names (axis 0) → unique observations
Column names (axis 1) → features, aka variables
and the data formed by their "product"

	studyName	Sample Number	Species	Region	Island	Stage	Individual ID	Clutch Completion	Date Egg	Culmen Length (mm)	Culmen Depth (mm)	Flipper Length (mm)	Body Mass (g)	Sex
row_id														
0	PAL0708	1	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N1A1	Yes	11/11/07	39.1	18.7	181.0	3750.0	MALE
1	PAL0708	2	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N1A2	Yes	11/11/07	39.5	17.4	186.0	3800.0	FEMALE
2	PAL0708	3	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N2A1	Yes	11/16/07	40.3	18.0	195.0	3250.0	FEMALE
3	PAL0708	5	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N3A1	Yes	11/16/07	36.7	19.3	193.0	3450.0	FEMALE
4	PAL0708	6	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N3A2	Yes	11/16/07	39.3	20.6	190.0	3650.0	MALE
5	PAL0708	7	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N4A1	No	11/15/07	38.9	17.8	181.0	3625.0	FEMALE
6	PAL0708	8	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N4A2	No	11/15/07	39.2	19.6	195.0	4675.0	MALE
7	PAL0708	13	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N7A1	Yes	11/15/07	41.1	17.6	182.0	3200.0	FEMALE
8	PAL0708	14	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N7A2	Yes	11/15/07	38.6	21.2	191.0	3800.0	MALE
9	PAL0708	15	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N8A1	Yes	11/16/07	34.6	21.1	198.0	4400.0	MALE

Reading along Axis 0

	studyName	Sample Number	Species	Region	Island	Stage	Individual ID	Clutch Completion	Date Egg	Culmen Length (mm)	Culmen Depth (mm)	Flipper Length (mm)	Body Mass (g)	Sex
row_id														
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3	PAL0708	5	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N3A1	Yes	11/16/07	36.7	19.3	193.0	3450.0	FEMALE
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9	PAL0708	15	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N8A1	Yes	11/16/07	34.6	21.1	198.0	4400.0	MALE

Reading along Axis 0

Looks at **one feature at a time**

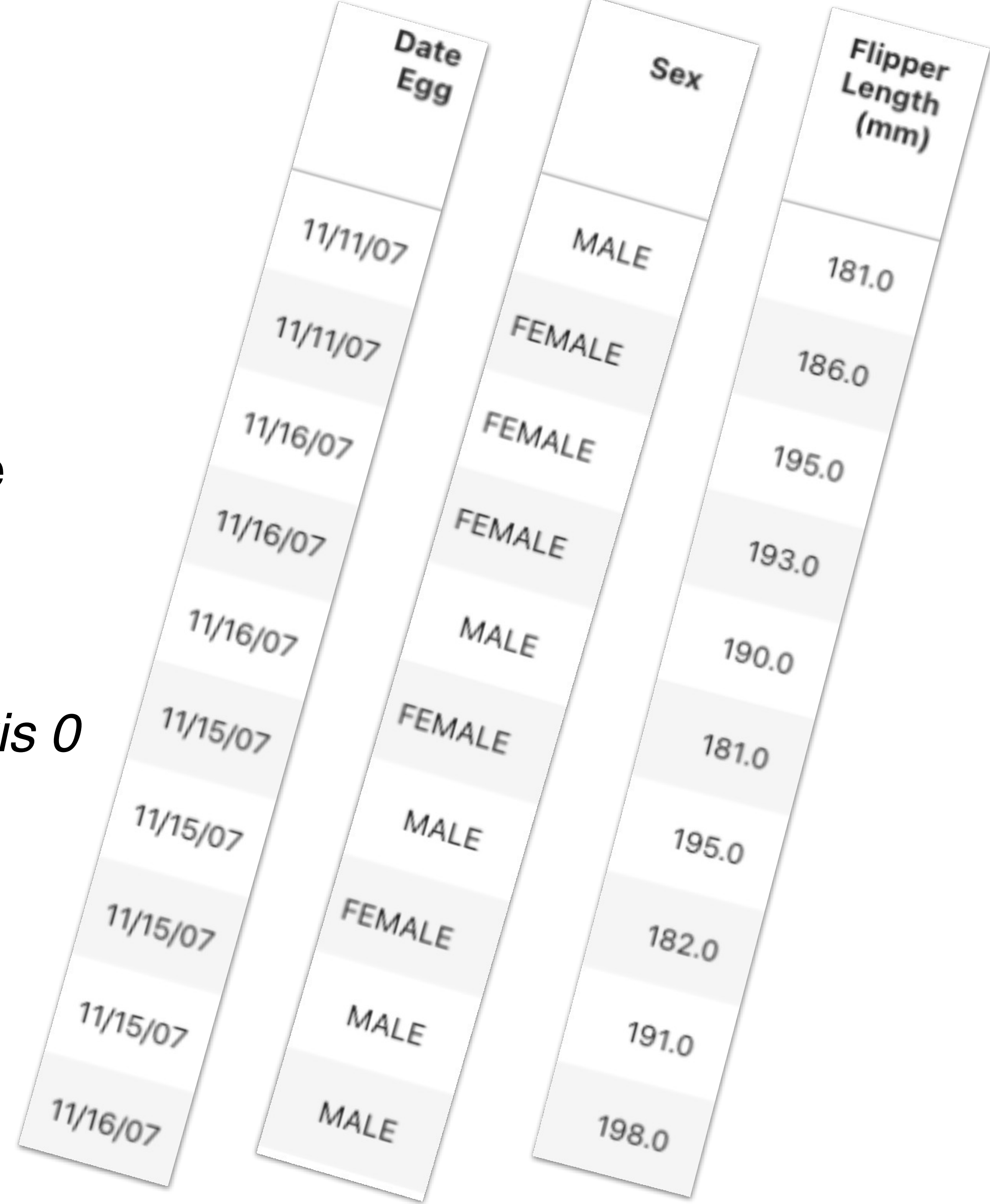
A feature is a variable

All **values** in the column are of the **same** type

All the values come from the "**domain**" of the variable

→ *Most visualizations pull data from along axis 0*

Ironically, called "column-wise"



Date Egg	Sex	Flipper Length (mm)
11/11/07	MALE	181.0
11/11/07	FEMALE	186.0
11/16/07	FEMALE	195.0
11/16/07	FEMALE	193.0
11/16/07	MALE	190.0
11/15/07	FEMALE	181.0
11/15/07	MALE	195.0
11/15/07	FEMALE	182.0
11/15/07	MALE	191.0
11/16/07	MALE	198.0

Reading along Axis 1

	studyName	Sample Number	Species	Region	Island	Stage	Individual ID	Clutch Completion	Date Egg	Culmen Length (mm)	Culmen Depth (mm)	Flipper Length (mm)	Body Mass (g)	Sex
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9	PAL0708	15	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N8A1	Yes	11/16/07	34.6	21.1	198.0	4400.0	MALE

Reading along Axis 1

- Looks at one observation at a time
- The values in the row may be of different types
- The values together represent a kind of signature
- Some visualizations pull data from along axis 1*
- "row-wise"

0	PAL0708	1	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N1A1	Yes	11/11/07	39.1	18.7	181.0	3750.0	MALE
1	PAL0708	2	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N1A2	Yes	11/11/07	39.5	17.4	186.0	3800.0	FEMALE
2	PAL0708	3	Adelie Penguin	Anvers	Torgersen	Adult, 1 Egg Stage	N2A1	Yes	11/16/07	40.3	18.0	195.0	3250.0	FEMALE

Tidy Data Tables

Wide vs. Long Data Table

Country	Medal Type	Count
United States	Gold	46
United States	Silver	29
United States	Bronze	29
China	Gold	38
China	Silver	27
China	Bronze	23

Table 1: Long Form

Each column is an attribute

Country	Gold	Silver	Bronze
United States	46	29	29
China	38	27	23

Table 2: Wide Form

Use attribute values as columns

“[Tidy Data](#)”, *Journal of Statistical Software*, by Hadley Wickham

Exercise

Discuss prompt in groups of 3

Year	Property_crime_rate
Reported crime in Alabama	
2004	4029.3
2005	3900
2006	3937
2007	3974.9
Reported crime in Alaska	
2004	3370.9
2005	3615
2006	3582
2007	3373.9

State	2004	2005	2006	2007
Alabama	4029.3	3900	3937	3974.9
Alaska	3370.9	3615	3582	3373.9

Which format would make it easier to compare trends over time?

Attribute Types

N	Nominal	Names and categories	N	Nominal	Names and categories	=, ≠
O	Ordinal	Ordered	O	Ordinal	Ordered	=, ≠, <, >
Q	Interval	Degree of difference	Q	Interval	Degree of difference	=, ≠, <, >, −
Q	Ratio	Unique and non-arbitrary zero value	Q	Ratio	Unique and non-arbitrary zero value	=, ≠, <, >, −, %
O	Ordinal	Ordered	N	Nominal	Names and categories	=, ≠
Q	Interval	Degree of difference	O	Ordinal	Ordered	=, ≠, <, >
Q	Ratio	Unique and non-arbitrary zero value	Q	Interval	Degree of difference	=, ≠, <, >, −
			Q	Ratio	Unique and non-arbitrary zero value	=, ≠, <, >, −, %
Q	Interval	Degree of difference	N	Nominal	Names and categories	=, ≠
Q	Ratio	Unique and non-arbitrary zero value	O	Ordinal	Ordered	=, ≠, <, >
			Q	Interval	Degree of difference	=, ≠, <, >, −
			Q	Ratio	Unique and non-arbitrary zero value	=, ≠, <, >, −, %

Attribute Types

Nominal

N	Nominal	Names, labels, and categories	<table><tr><td>N</td><td>Nominal</td><td>Names, labels, and categories</td><td>=, ≠</td></tr><tr><td>O</td><td>Ordinal</td><td>Ordered</td><td>=, ≠, <, ></td></tr><tr><td>Q</td><td>Interval</td><td>Degree of difference</td><td>=, ≠, <, >, −</td></tr><tr><td>Q</td><td>Ratio</td><td>Unique and non-arbitrary zero value</td><td>=, ≠, <, >, −, %</td></tr></table>	N	Nominal	Names, labels, and categories	=, ≠	O	Ordinal	Ordered	=, ≠, <, >	Q	Interval	Degree of difference	=, ≠, <, >, −	Q	Ratio	Unique and non-arbitrary zero value	=, ≠, <, >, −, %
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Q	Interval	Degree of difference	=, ≠, <, >, −																
Q	Ratio	Unique and non-arbitrary zero value	=, ≠, <, >, −, %																

Nominal

Fruits: apples, oranges, ..

Attribute Types

Ordinal

N	Nominal	Names and categories	<table><tr><td>N</td><td>Nominal</td><td>Names and categories</td><td>=, ≠</td></tr><tr><td>O</td><td>Ordinal</td><td>Ordered</td><td>=, ≠, <, ></td></tr><tr><td>Q</td><td>Interval</td><td>Degree of difference</td><td>=, ≠, <, >, −</td></tr><tr><td>Q</td><td>Ratio</td><td>Unique and non-arbitrary zero value</td><td>=, ≠, <, >, −, %</td></tr></table>	N	Nominal	Names and categories	=, ≠	O	Ordinal	Ordered	=, ≠, <, >	Q	Interval	Degree of difference	=, ≠, <, >, −	Q	Ratio	Unique and non-arbitrary zero value	=, ≠, <, >, −, %
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Ordinal

Quality of meat (Grades): A, AA, AAA

Attribute Types

Q-Interval

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Q-interval

Dates: Jan 19, 2006

Location: Lat 33.98, long -118.45

Only differences can be calculated (e.g., distances or spans)

Attribute Types

Q-Ratio

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Q-Ratio (fixed zero)
Physical measurements: Length, Mass, ...

Includes counts and amounts
Can measure ratios or proportions

Don't ask “is there a zero” but instead ask “does zero mean the absence of X?”
e.g. There is a “year 0” in our calendar system but it does *not* mean “absence of years”

NOIR Scramble

Step 1: figure out what type of data you have

Step 2: Find three other students who have the other three attribute types to make a **NOIR group of four**

Step 3: Check in with me; if you are right, write your attributes on the board!



Takeaways for design

1. Always interrogate your data first:

Is the data frame tidy?

Is it in wide or long format?

What is an observation? What are the relationships/properties?

2. For each column, what is the attribute type?

We will talk more about what encodings/plots are possible and effective for each attribute type

3. Should the data be grouped in some way?

4. Are there particular relationships, attributes, or trends you want to highlight with visualization?

What's next

- In the absence of a quiz today, write down your answer to the following question: **If you had to explain today's concepts to a friend, which would be hardest for you to explain?**
- **Friday's lab** will explore these concepts using pandas data frames
- Next week:
 - Read "**A Tour through the Visualization Zoo**"
 - Quiz** on this and previous material on Wednesday
 - Your topic and dataset checkpoint for Project 1 is **due Sep 13**